Summary

OO Modeling Process.

- Conceptual modeling captures ideas and concepts so they so they came be analyzed, prioritized and designed. It's purpose are to understand real world-systems, visualize the current and desired system, document key decisions, specify asolution, plan construction and, above all, comunicate efficiently.

- UML is a collection of modeling languajes: use case drograms dass drograms, interaction drograms and state charts.

- Good practices

1. Loosely coupled layers; each layer offers a clear abstraction to higher layers and many use its own layer and the one below.
2. Prefer aggregation over inheritance; aggregation can change at runtime; so it is more flexible.
3. Program to an interface or abstraction this reduces coupling and unnecesary dependences.
4. Meaningful identifers; classes are nouns is PascalCase variables lowerCamelCase and reflect their purpose.

Introduction to UML

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uml is an object-oriented modeling language that is expressive semiformal, supports incremental development, is a process independt, and is customizable and extensible

- Buiding blocks; modeling elements are like words, diagrams like sentences or paragraphs and models like whole papers or books

- Diagram types
  - Behavioral: use case, sequence, communication, state chart
  - Structural: class, object, component, deployment.

Three modeling perspectives.

- Analisys; real-world objects, the problem domain, the system as it is.
- Specification; real-world and software objects, automaton boundaries, what the system is to become.
- Implementation; mostly software objects, solution-oriented concepts

Basic Use Cases

1. Actors: things outside the system that have goals with respect to it. They are named in the singular and

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draw as stick figures. All users are actors, but not all actors are users.

2. Use cases: what users need to accomplish, indepent of design choises.

3. Use cases v.s system interactions: use cases answer what and why; interactions answer how. Confusing them leads to poor understanding of why features exist.

4. Use case drogram: actors, use casses, relationships and subjects.

5. Relationships between use cases:
   - 5.1. extend: a variation of anoter use case under condition, at an extension point. The extended use case stands me.
   - include: one uses requires another's goal
   - 5.3. Generalization: one use case is a subject or refinent

6. Finding actors: ask who or what uses the system, provides input, usses output, needs suport.

7. Finding use casses: list actors and their activities, or list external events and the system's response.

Objects, Classes and Basic Class Diagrams.

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Objects

Objects are the basic building blocks of object oriented systems. An object represents any "thing", has a boundary in space and/or time, is an abstraction, encapsulates and behavior, and has identity

Usually not objects: events, relationships between objects behavior, and constraints

Classes

Systems contain many objects that come and go. Humans classify similar things by their observable properties and classes are abstraction that let us handle whole collections of objects that share commonalities.

Example: student, Course, Semester, Instructor, Department college, Class section, Major, Grade.

UML Quick Reference Guide

General best practices Keep an analysis perspective let the incompleteness of diagrams drive questions treat the model as an essay and each diagram as a

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paragraph with one central idea, pick the right diagram type, avoid line crossings and use color effectively

- Use case diagram; actor, use case a solid line without arrowheads between actor and use case, extend and notes.
- Class diagrams; class, generalization, named binary association, aggregation, composition, n-ary association multiplicity and attributes.
- Activity diagrams; initial node, action, control flow decision, fork/join, object flow, send/accept signal and final node.
- State chars: initial state, transition, internal activity highleved states, internal transiction, entry/exit actions and final state.

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